

Solution: From Theorem 10, we know that this statement is True.
b) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotates vectors about the origin through an angle θ , then T is a linear transformation.

Solution: From Example 3 on page 73, we know that the given statement is True.

c) When two linear transformations are performed one after another, the combined effect may not always be a linear transformation.

Solution: On page 73, in the last sentence of the section "Geometric Transformations of $\mathbb{R}^2$ ", it is mentioned that Section 2.1 will show that a composition of linear transformations is linear. $\therefore$ The given statement is False.

d) A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is onto $\mathbb{R}^m$ if every vector z in $\mathbb{R}^m$ maps onto some vector in $\mathbb{R}^m$. By definition of onto on page 76, we know that the given statement is False.

Solution: By definition of onto on page 76, we know that the given statement is False.

e) If A is a 3×2 matrix, then the transformation $x \mapsto Ax$ cannot be one-to-one.

Solution: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. It's very easy to prove that the columns of A are not multiples of each other. $\therefore$ The columns of A are linearly independent. By Theorem 12(b), the transformation $x \mapsto Ax$ is one-to-one. $\therefore$ The given statement is False. $\square$

2a) Not every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a matrix transformation. $\therefore$ The given statement is False. $\square$

Solution: From Theorem 10, we know that the given statement is False.
b) The columns of the standard matrix for a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ are the images of the columns of the $n \times n$ identity matrix.

Solution: By the defn. of image of x on page 64 and Theorem 10, we know that the given statement is True.